Chapter

4

Organic Chemistry

5.2.8 (2015:14)

1.

Which one of the following lists the solubilities of butane (C_4H_{10}), butan-2-ol ($CH_3CH(OH)CH_2CH_3$) and butanone ($CH_3COCH_2CH_3$) in water, from **most** soluble to **least** soluble?

(a)	butan-2-ol	butanone	butane
(b)	butan-2-ol	butane	butanone
(c)	butanone	butan-2-ol	butane
(d)	butane	butanone	butan-2-ol

5.2.8 (2015:22)

2.

Which one of the following lists the substances in order of **increasing** (from lowest to highest) boiling point?

(a)	CH_3CH_3	CH_3CH_2OH	CH_3CHO	CH_3COOH
(b)	CH_3CH_3	CH_3CHO	CH_3CH_2OH	CH_3COOH
(c)	CH_3CH_2OH	CH_3CH_3	CH_3COOH	CH_3CHO
(d)	CH_3COOH	CH_3CHO	CH_3CH_2OH	CH_3CH_3

5.2.10 (2015:23)

3.

Under the right conditions, a compound containing two double bonds, buta-1,3-diene ($H_2C{=}CH{-}HC{=}CH_2$), can react with itself to make Buna rubber. This process is **best** referred to as

(a) saponification.

(b) condensation polymerisation.

(c) esterification.

(d) addition polymerisation.

5.2.3 (2015:24)

4.

What is the name of the organic compound produced when 2-fluoropent-1-ene reacts with fluorine gas?

(a) 2-fluoropentane

(b) 1,2-difluoropentane

(c) 1,1,2-trifluoropentane

(d) 1,2,2-trifluoropentane

5. *5.2.3 (2015:25)*

Between which of the following pairs of substances can hydrogen bonding occur?

I	CH_3COCH_3 and CH_3NH_2
II	CH_3CHO and HF
III	C_2H_6 and CH_3OH
IV	CH_3F and H_2O

(a) I, II and III only

(b) I, II only

(c) I, III and IV only

(d) II only

6. *5.2.6 (2016 SP:17)*

Which one of the following will react with acidified potassium dichromate solution to give a ketone?

(a) $CH_3CH_2CH_2OH$

(b) CH_3CH_2CHO

(c) $CH_3CH(OH)CH_3$

(d) $(CH_3)_3COH$

7. *5.2.8 (2016 SP:20)*

Consider the following substances.

(i) $BaSO_4$ (ii) $CH_3CH_2CH_2CH_2OH$

(iii) $CH_3CH_2COCH_3$ (iv) H_2NCH_2COOH

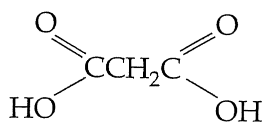
Which one of the following lists the substances in order of **decreasing** solubility in water?

(a)	i	iv	ii	iii
(b)	i	iii	ii	iv
(c)	iv	ii	iii	i
(d)	ii	iv	iii	i

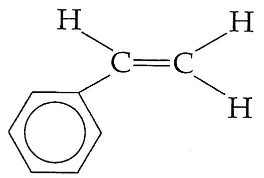
8. *5.2.13 (2016 SP:23)*

Which one of the following pairs represent monomers that could react together to form a polymer?

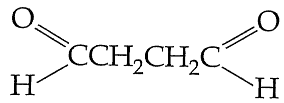
(i)



(ii)



(iii)



(iv)

$HO—CH_2CH_2CH_2CH_2—OH$

(a) i and iv

(b) i and iii

(c) ii and iii

(d) iii and iv

9. *5.1 (2016 SP:24)*

Proteins that show a high degree of similarity in their primary structure in the Protein Data Bank are most likely to have

(a) similar function.

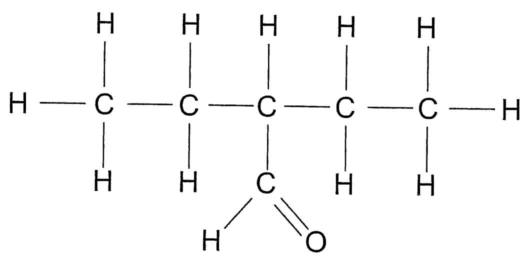
(b) identical tertiary structure.

(c) been isolated from the same species.

(d) the same amino acid composition.

10. *5.2.4 (2016:18)*

What is the IUPAC name of the following compound?



(a) 3-methylpentan-3-al

(b) 2-ethylbutanal

(c) 2,2-diethylethanal

(d) 2-methylbutanal

5.2.6 (2016:19)

11.

Below is a table of reactions involving organic compounds.

Reaction	Product
ethene + hydrogen	1
ethanal + acidified permanganate solution	2
ethanol + acetic (ethanoic) acid with sulfuric acid	3
acetic (ethanoic) acid + sodium carbonate	4

Which row of the table below identifies a product of each reaction correctly?

	Product 1	Product 2	Product 3	Product 4
(a)	an alkane	a carboxylic acid	an aldehyde	an ester
(b)	an alkane	a carboxylic acid	an ester	carbon dioxide
(c)	an alkene	carbon dioxide	an aldehyde	a carboxylic acid
(d)	an alkane	carbon dioxide	an ester	a carboxylic acid

5.2.10 (2016:20)

12.

Which of the following compounds could be used to produce a polymer?

I CH_2CHCH_3
II $HOOCCH_2COOH$
III CH_2CHOH
IV $HOCH_2CH_3$
V $H_2NCH_2NH_2$

(a) I, II, V only

(b) I, II, IV only

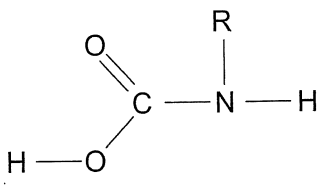
(c) I, II, III, V only

(d) II, III, IV, V only

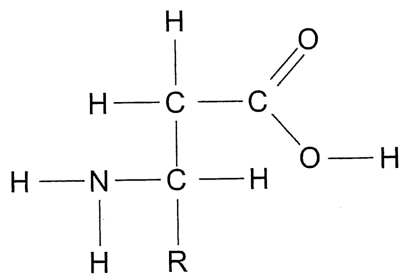
13. *5.2.14 (2016:21)*

Which of the following **best** represents the generalised structure of α-amino acids? (Note: R represents a side chain.)

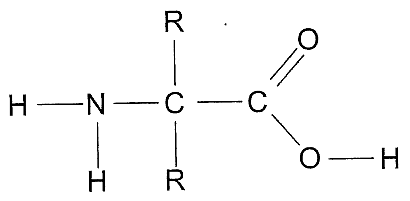
(a)



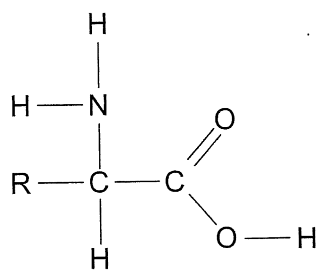
(b)



(c)



(d)



14. *5.2.5 (2017:14)*

Which of the following are isomers of $C_5H_8O_2$?

(i) $CH_3CH_2COCH_2CHO$

(ii) $CH_3CH_2CH_2CH_2COOH$

(iii) $CH_3COCH(CH_3)CHO$

(iv) $CH_2CHCH_2CH_2COOH$

(a) i and ii only

(b) i, ii and iv only

(c) i, iii and iv only

(d) ii, iii and iv only

15. *5.2.14 (2017:15)*

Which one of the following is the dominant form of glycine in basic solution?

(a) NH_2-CH_2-$COOH$

(b) NH_2-CH_2-COO^-

(c) NH_3^+-CH_2-COO^-

(d) NH_3^+-CH_2-$COOH$

16. *5.2.3 (2017:16)*

A chemist attempts to identify a pungent, colourless liquid by conducting several experiments. The results are shown in the table below:

Experiment	Observations
add acidified potassium dichromate solution	orange solution turns green
a lighted taper held above the liquid	flame and heat produced
add sodium metal	metal reacts and colourless, odourless gas evolved
add acidified, concentrated acetic (ethanoic) acid	fruity odour produced

Using this information, identify the functional group present in the liquid.

(a) ketone

(b) alcohol

(c) amine

(d) carboxylic acid

17. *5.2.17 (2017:17)*

The amino acid sequence of a protein is referred to as its

(a) primary structure.

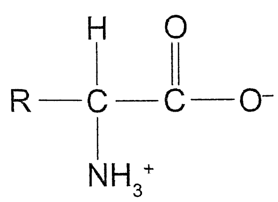
(b) secondary structure.

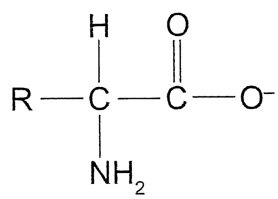
(c) tertiary structure.

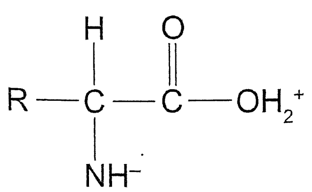
(d) parent chain.

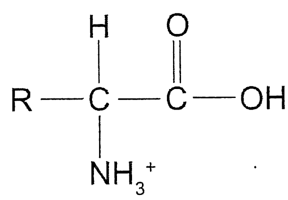
18. *5.2.15 (2017:19)*

Which one of the following structures represents a zwitterion?

(a) 

(c) 

(b) 

(d) 

19. *5.1 (2017:20)*

The function of a protein is linked closely to

(a) its method of production.

(b) the nature of its intermolecular forces.

(c) the number of atoms bonded to it.

(d) its structure.

5.2.7 (2018:04)

20.

The compound with the structural formula shown below smells like apricots:

$$CH_3-CH_2-CH_2-CH_2-CH_2-O-\overset{\overset{\displaystyle O}{||}}{C}-CH_2-CH_2-CH_3$$

Which of the following is true for this compound?

	Name of compound	Organic reactants required to synthesise this compound
(a)	pentyl butanoate	pentanol and butanoic acid
(b)	butyl pentanoate	butanol and pentanoic acid
(c)	pentyl butanoate	butanol and pentanoic acid
(d)	butyl pentanoate	pentanol and butanoic acid

5.1 (2018:09)

21.

Which of the following statements about the Protein Databank (PDB) is/are correct?

(i) The PDB allows users to select a protein and then view its structure.

(ii) The PDB is updated regularly and access by scientists worldwide is free.

(iii) The PDB is a worldwide repository of information on all chemical substances listed in chronological order of discovery.

(a) i only

(b) iii only

(c) i and ii only

(d) ii and iii only

5.2.5 (2018:11)

22.

Which of the following molecules is capable of demonstrating *cis-trans* isomerisation?

(a) $CH_2CHCHBrCH_3$

(b) $CH_3CHCHCH_3$

(c) $CBr_2CHCH_2CH_2Br$

(d) $CH_2BrCBr_2CH_2CH_3$

5.2.18 (2018:17)

23.

Proteins can contain α-helices and/or β-pleated sheets. The intermolecular forces holding these structures in their shapes are

(a) dispersion forces.

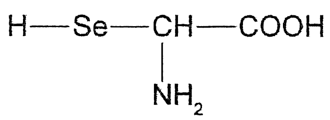
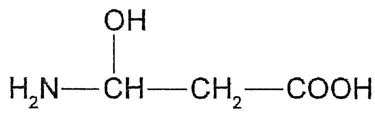
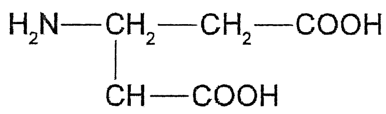
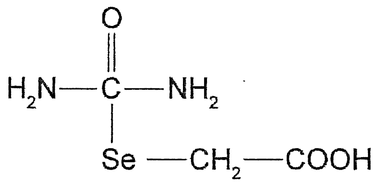
(b) dipole-dipole forces.

(c) hydrogen bonds.

(d) ion-dipole attractions.

24. *5.2.14 (2019:09)*

Which one of the following is an alpha amino acid?

(a)		(b)	
(c)		(d)	

25. *5.2.6 (2019:14)*

Which one of the following properties exhibited by octanol is **not** related to the dispersion forces between the molecules?

(a) combustibility

(b) melting point

(c) solubility in octane

(d) solubility in water

26. *5.2.5 (2019:15)*

Which one of the following compounds will **not** exhibit *cis-trans* isomerism?

(a) 1,2-difluoro-but-1-ene

(b) 1,1-difluoro-but-1-ene

(c) 1,2-difluoro-but-2-ene

(d) 1,4-difluoro-but-2-ene

27. *5.2.6 (2019:18)*

Which one of the following could **not** be a product when propan-1-ol is oxidised?

(a) CO_2

(b) CH_3CH_2CHO

(c) CH_3CH_2COOH

(d) CH_3COCH_3

28. *5.2.8 (2019:22)*

Between which of the following pairs of substances can dispersion forces exist?

(i) $CH_3C\ell$ and H_2O

(ii) CH_3CH_2CHO and HBr

(iii) $CH_3CH_2CH_2CH_3$ and $CH_3CH_2CH_2CH_2CH_2CH_3$

(iv) $CH_3CH_2CH_2OH$ and NH_3

(a) i and ii only

(b) i, ii and iii only

(c) iii only

(d) i, ii, iii and iv

29. *5.2.5 (2019:23)*

Which one of the following is an isomer of pentanoic acid?

(a) $CH_3CHCH\text{-}O\text{-}CH_2CHO$

(b) $CH_2CHCH_2\text{-}O\text{-}CH_2CH_2OH$

(c) $OHCCH_2CH_2CH_2CHO$

(d) $CH_3CHCHCH_2COOH$

30. *5.2.5 (2019:25)*

How many isomers does the compound $C_2H_3Br_3$ have?

(a) 1

(b) 2

(c) 3

(d) 4

31. *5.2.5 (2020:04)*

The number of possible isomers of $C_2H_2F_2$ is

(a) 1

(b) 2

(c) 3

(d) 4

5.2.1 (2020:10)

32.

Which of these statements regarding organic molecules are correct?

(i) Organic molecules all contain carbon.

(ii) Functional groups consist of groups of atoms or a particular type of bond.

(iii) Functional groups influence the chemical properties of organic molecules.

(iv) Functional groups influence the physical properties of organic molecules.

(a) i and iii only

(b) ii and iv only

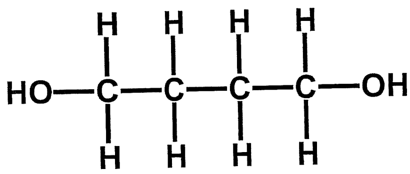
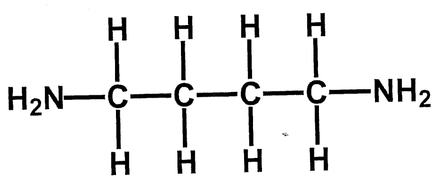
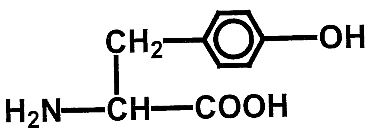
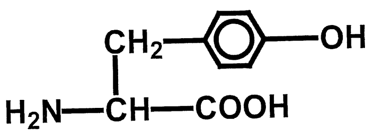
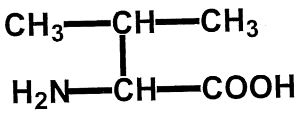
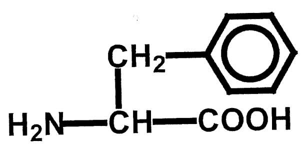
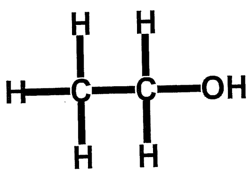
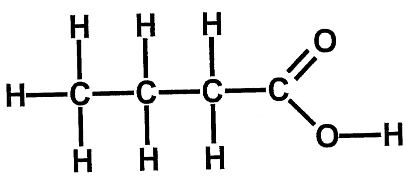
(c) i, ii and iii only

(d) i, ii, iii and iv

5.2.15 (2020:11)

33.

Which of the following pairs of molecules can form peptide bonds with each other?

(i)	 butane-1,4-diol	 butane-1,4-diamine
(ii)	 tyrosine	 tyrosine
(iii)	 valine	 phenylalanine
(iv)	 ethanol	 butanoic acid

(a) i and iv only

(b) ii and iii only

(c) i, ii and iii only

(d) i, ii, iii and iv

34. *5.2.8 (2020:13)*

Which of the following alcohols would you expect to have the highest boiling point?

(a) pentan–1–ol

(b) pentan–2–ol

(c) pentan–3–ol

(d) 2–methylbutan–2–ol

35. *5.1 (2020:14)*

The Protein Data Bank contains information relating to the structures of proteins. The structure of a protein is important because it is related closely to its

(a) equilibrium constant.

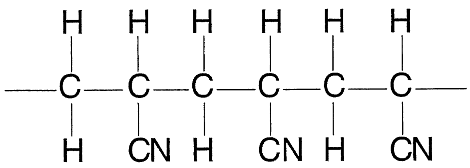
(b) bonding capacity.

(c) nutritional value.

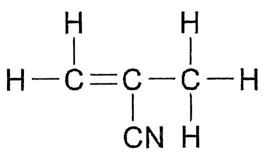
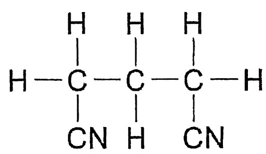
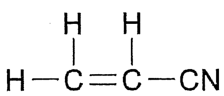
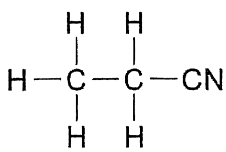
(d) function.

36. *5.2.11 (2020:21)*

Polyacrylonitrile fibres can be used to make blankets and carpets. The structural formula of a segment of this polymer is shown below.



The structural formula of the monomer used to make polyacrylonitrile is:

(a)		(b)	
(c)		(d)	

7.(2016 SP:16-d) A fuel cell is a REDOX cell utilising gases. So answer (i) is correct. Answer (ii) is correct. As the main product of hydrogen fuel cells is water vapour it is a low-emission technology. They cannot be recharged by reversing the flow of electricity. The answer is 'd'.

8.(2016:13-d) Calculation of oxidation numbers is specifically mentioned in the syllabus. The negative ON is awarded to the element closest to fluorine. So for nitrogen, Step 1: $0 \rightarrow -3$, Step 2: $-3 \rightarrow +2$, Step 3: $+2 \rightarrow +4$, Step 4: $+4 \rightarrow +5$ a change of 1. The answer is 'd'.

9.(2016:14-a) Students might see chromate and dichromate and assume it is a redox reaction in part a. It is in fact acid/ base. In all other cases the ON of chromium changes. The answer is 'a'.

10.(2016:15-c) This question can be solved using the Data Book. Reducing agents get oxidised so we need to look at equations where the substance is losing electrons. Check the E° values of the reactions making sure you have them all as oxidation reactions. The answer is 'c'.

11.(2016:16-c) This question can be solved using the Data Book. Find the E° values of each of the half-equations. Add them and find which total is negative – i.e. it is NOT spontaneous. The answer is 'c'.

12.(2016:17-a) The element C reduced both A^{2+} and B^{2+}. Thus it is the most powerful reducing agent. A reduced B^{2+} so the answer is 'a'.

13.(2017:02-b) An exercise in using the oxidation number rules that you should know. The answer is 'b'.

14.(2017:03-b) This question involves the understanding that for a fixed quantity of electrons the valency of an ion will determine how many mol of product will form. The answer is 'b'.

15.(2017:04-c) This question appeared in the 2016 examination as well. It recognises that both galvanic and electrolytic cells have external wires, electrodes and redox reactions. The answer is 'c'.

16.(2017:09-c) Corrosion is a redox process and this question is essentially asking which of these is not a redox process. Iron(III) chloride dissolving is not redox. The answer is 'c'.

17.(2017:10-a) Before wasting time in calculating the E° for every equation perhaps stop and think. In (d) iodine cannot displace chlorine (it is further down the group). In (c) silver ion and bromide ion will not form elements. In (b) gold does not rust we know that. Then using E° check (a). The answer is 'a'.

18.(2018:05-a) The question asks for the E° for the reaction $In^{3+}(aq) + 3\,e^- \rightarrow In(s)$. In the question the dichromate reaction was +1.36 V and the total was +1.70 V so the $In \rightarrow In^{3+} + 3\,e^-$ reaction must have been +0.34 V. This question asks for the reverse equation so the E° is – 0.34 V.

19.(2018:06-d) The question is actually asking 'which metals will have a negative E° for the following reaction $In^{3+}(aq) + M(s) \rightarrow In(s) + M^{+}(aq)$?' The reaction $M \rightarrow M^{x+} + x\,e^-$ must have an E° < + 0.34 V as the $In^{3+}(aq) + 3\,e^- \rightarrow In(s)$ reaction is –0.34 V. Ni = + 0.24 V, Sn = +0.14 V, Cu = –0.34 V, Ag = –0.80 V. The answer is 'd'.

20.(2018:07-d) Oxidation numbers are: MoO_2(+4), $Mo_2O_7^{2-}$(+6), $HMoO_4^{2-}$(+5). The answer is 'd'.

21.(2018:14-c) Note Zr^{4+} is not reduced by any of the metals. This means Zr is a very active metal and so is most easily oxidised. The answer is 'c'.

22.(2018:15-a) A reducing agent gets oxidised (loses electrons) and forces another substance to be reduced (accept electrons). Rhenium metal fails to reduce any ions and is therefore the weakest. The answer is 'a'.

23.(2018:25-a) In all electrochemical cells (galvanic and electrolytic) ions migrate and oxidation occurs at the anode. The answer is 'a'.

24.(2019:06-c) This question involved selecting the appropriate half equation using oxygen and reversing the lead equation. The answer is 'c'.

25.(2019:10-d) This question could be done quickly by noting the relative activity of Group 17 elements or it could be done slowly by calculating the E° values. The answer is 'd'.

26.(2019:17-b) The quickest way to approach this problem is to realise elements have an oxidation number of zero. This eliminates (i) and (iv) – then by calculation the answer is 'b'.

27.(2020:01) Ho(s) has changed oxidation number from 0 to +3 ∴ it has been oxidised so it is a reducing agent. Hydrogen in water has changed from +1 to 0 in $H_2(g)$ ∴ it has been reduced so it is an oxidising agent. Answer is (c).

28.(2020:03) A REDOX definition question – the answer is (b).

29.(2020:06) Galvanic cells are spontaneous – they produce energy in the form of electricity. The answer is (d).

30.(2020:07) This question is based on the fact concentrated nitric acid can be an oxidising agent as well as being an acid. In this case it is acting as an oxidising agent and copper is oxidised. (i) is true, (ii) is not true (iii) copper is oxidised. The answer is (a).

31.(2020:19) A reducing agent reduces other substances – that is, it donates electrons to them. So we are talking about the reverse reactions. The strongest reverse reaction will be the third with E° of +1.34. The answer is (d).

32.(2020:20) The sum of the E° values is negative – this cell is not spontaneous – it must be forced. Therefore, it is an electrolytic cell. The answer is (a).

Chapter 4: Organic Chemistry

1.(2015:14-a) Answer 'a'. Hydrogen bonding causes more solubility in water than dispersion forces.
2.(2015:22-b) Answer 'b'. Dispersion forces < hydrogen bonding < hydrogen bonding and dipole bonding.
3.(2015:23-d) Answer 'd'. Double bonds engage in addition polymerisation.
4.(2015:24-d) Answer 'd'. Addition is always across the double bond.
5.(2015:25-b) Answer 'b'. III cannot accept or donate H bonds.
6.(2016 SP:17-c) Secondary alcohols can be oxidised to form ketones. $CH_3CH(OH)CH_3$ is a secondary alcohol. The answer is 'c'.

7.(2016 SP:20-c) Barium sulfate is insoluble. Number (iv) can donate hydrogen bonds and accept them in two places so will be very soluble. Number (ii) can accept and donate hydrogen bonds so will be quite soluble. The answer is 'c'.

8.(2016 SP:23-a) To form a polymer we need a link to be formed between monomers. Substance (a) has carboxylic acid groups and could form ester linkages. So with (iv) this could form a polymer. (ii) and (iii) cannot form linkages. The answer is 'a'.

9.(2016 SP:24-a) The primary structure of a protein is the covalently bonded polypeptide polymer. Its 'R' groups are determined by the amino acids in the polymer. The functional groups in the 'R' groups determine the tertiary structure. If two polymers have similar functional groups they will have similar tertiary structures and hence similar shape. Function is dependent on shape. The answer is 'a'.
The strength of the interaction between ethanol molecules and water molecules are stronger than the internal interactions of both so ethanol dissolves in water – therefore they are miscible.

10.(2016:18-b) The –CH = O group is an aldehyde. Aldehydes always appear at the end of a chain. The base substance is therefore butanal. The answer is 'b'.

11.(2016:19-d) The reactions are addition to an alkene, oxidation of an aldehyde, esterification, and neutralisation acid with a carbonate. The answer is 'd'.

12.(2016:20-c) To make a polymer either an alkene is needed (to make an additional polymer) or the necessary groups to form a condensation polymer (polyester or polyamide). Addition is possible with I and III, while a polyester can form with II and V. The answer is 'c'.

13.(2016:21-d) An α-amino acid has a central α carbon a COOH, one side group (or an H) and an NH_2. In 'a' there is no central C, 'b' has two central carbons, 'c' has two groups. The answer is 'd'.

14.(2017:14-c) To be isomers the molecules must have the same molecular formula. The answer is 'c'.

15.(2017:15-b) The question checks recognition of the fact that an amino acid contains a COOH acid group that will react with a strong base to form COO^-. The answer is 'b'.

16.(2017:16-b) The syllabus requires knowledge of the chemical properties of functional groups and 'these reactions can be used to identify the functional group present within the organic compound'. In turn the tests show it can be oxidised, it is flammable, it has an OH group and it is either a carboxylic acid or an alcohol. The answer is 'b'.

17.(2017:17-a) A question straight from the syllabus: 'the sequence of α-amino acids in a protein is called its primary structure.' The answer is 'a'.

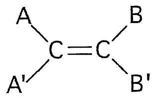
18.(2017:19-a) This question is taken directly from the syllabus and identifies the acid and base properties of amino acids. The NH_2 accepts a proton and the COOH loses one. The answer is 'a'

19.(2017:20-d) This question is taken directly from the syllabus. The 'lock and key' model of protein function means the shape is the essential characteristic in protein function and this is due to the structure or shape. The answer is 'd'.

20.(2018:04-a) When naming an ester look for the double bonded oxygen. This was part of the original carboxylic acid. The left-hand part was therefore the alcohol. So butanoic acid and pentan-1-ol were used. The substance is pentyl butanoate. The answer is 'a'.

21.(2018:09-c) The PDB allows users to access the structures of proteins and view them in 3D 'cartoon' mode. It is free of charge and lists proteins alphabetically. The answer is 'c'.

22.(2018:11-b) In an alkene cis-trans isomerism (not called geometric isomerism anymore) can occur when A and A' are different and when B and B' are different also.



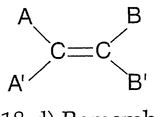
This only occurs in case b. The answer is 'b'.

23.(2018:17-c) The secondary structure is formed by hydrogen bonding between the H in a N–H and an O in a C=O group. Depending on the amino acid sequence, it forms either α-helices and/or β-pleated sheets. The answer is 'c'.

24.(2019:09-a) An alpha amino acid has an N terminus, a C terminus and one central C called the alpha carbon. There is no requirement that the correct answer be amongst those listed on the data sheet. The answer is 'a'.

25.(2019:14-a) Melting point and solubility are related to secondary forces. Combustion concerns breaking covalent bonds to produce activated particles. The answer is 'a'.

26.(2019:15-b) To exhibit cis-trans isomerism molecules must have A and A' different **and** B and B' different.

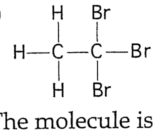
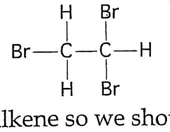


– The answer is 'b'.

27.(2019:18-d) Remember that combustion is a form of oxidation and that primary alcohols can be oxidised to aldehydes and/or carboxylic acids. The answer is 'd'.

28.(2019:22-d) Dispersion forces are created by instantaneous dipoles caused by electrons. By definition this means all of the substances have dispersion forces. The answer is 'd'.

29.(2019:23-b) Pentanoic acid (C_4H_9COOH) has a molecular formula $C_5H_{10}O_2$. Only one distractor has H_{10} – the answer is 'b'.

30.(2019:25-b)  and  The answer is 'b'.

31.(2020:04) The molecule is an alkene so we should consider cis–trans isomers too.
1,1–difluoroethene, cis–1,2–difluoroethene, trans–1,2–difluoroethane. Answer is (c).

32.(2020:10) Organic molecules contain carbon. Functional groups affect the chemical and physical properties of a molecule – by e.g. alcohol demonstrating hydrogen bonds that affects boiling point and the type of alcohol determine oxidation outcomes (aldehyde, ketone). The presence of double bonds similarly changes chemical and physical properties. The answer is (d).

33.(2020:11) A peptide (or amide) is a condensation product of an amine and a carboxylic acid that can be used to form a polymer. The constituent molecules should have suitable functional groups either $-NH_2$ or $-COOH$ must be present. The answer is (b).

34.(2020:13) All of these alcohols can demonstrate hydrogen bonding with each other. Pentan-1-ol molecules can pack closest together. Therefore, they will be the hardest to separate requiring more energy that is available at a high temperature. The answer is (a).

35.(2020:14) The structure (shape) of a protein determines its function like a key in a lock. The answer is (d).

36.(2020:21) This polymer is an example of an addition polymer. The repeating unit is $CH_2{=}CH{-}CN$. The answer is (c).

Chapter 5: Synthesis

1.(2016 SP:22-b) To form an ester an alcohol and a carboxylic acid are required. The ethene can form ethanol by an addition reaction in the presence of water, and the ethanol can be oxidised to form ethanoic acid. These two substances can react to form ethyl ethanoate. The answer is 'b'.

2.(2016:05-b) This is an equation mentioned in syllabus and should be known. NaOH produces soap while methanol produces the transesterification reaction making biodiesel. The answer is 'b'.

3.(2016:22-a) Hardness is a measure of calcium ion concentration. This is measured by volume of soap reacted. Other salts (chlorides etc.) will respond to silver ions or raise the boiling point but they have nothing to do with hardness. Only soap solution is indicative of hardness. The answer is 'a'.

4.(2016:23-b) Reaction X is a REDOX reaction with catalyst but this is not one of the answers offered. The best answer is 'b'.

5.(2016:24-c) This is a known reaction from the syllabus. The answer is 'c'.

6.(2016:25-a) This is a known reaction from the syllabus. The answer is 'a'.

7.(2017:11-d) There are 12 principles of green chemistry you should have read. The minimising toxic waste and by-products is essential. The answer is 'd'.

8.(2017:13-c) Rate determining steps have not been examined previously in the WACE. The idea is that in a sequence of reactions the overall reaction rate is dependent on the slowest step. The answer is 'c'.

9.(2017:21-c) The syllabus states 'the structure of the anionic detergents derived from dodecylbenzene contains a non-polar hydrocarbon chain and a sulfonate group'. You should know what a sulfonate group is. The answer is 'c'.

10.(2018:19-b) An enzyme lowers the activation energy so (a) and (c) are wrong. The reaction is exothermic so the answer is 'b'.

11.(2018:20-c) Answers (a), (b) and (d) are all **correct**. The hydration reaction does not use enzymes. The answer is 'c'.

12.(2018:21-b) The reverse reaction has a large Ea due to the fact the reaction is exothermic. Thus, the reverse reaction benefits more from the use of a catalyst than the forward reaction which has a relatively small Ea. The answer is 'b'.

13.(2018:22-d) The question asks only about yield. The reaction is exothermic thus removing heat will favour the forward reaction and the mol ratio indicates high pressure will also favour the forward reaction. The answer is 'd'.

14.(2018:23-a) Both reactions are exothermic and the Haber process uses a catalyst mostly made of iron while the Contact process uses vanadium pentoxide as a catalyst but remember catalysts do not alter yield – just rate. The answer is 'a'.

15.(2018:24-a) The long hydrocarbon chain demonstrates substantial dispersion forces with oil (another molecule with high dispersion forces). The long carbon chain has no lone pairs of electrons and so cannot engage in hydrogen bonding and since it has no ions or dipoles the answer is 'a'.

16.(2019:11-a) This question required detailed understanding of the two biodiesel processes and soap making along with some green chemistry. The answer is 'a'.

17.(2019:13-c) This question again required knowledge of the biodiesel process and required recognition of a triglyceride – (a) and (b) are products of the reaction. The answer is 'c'.

18.(2020:09) This is the standard structure of an alkyl (the straight chain) benzyl (the benzene ring) sulfonate (the negatively charged SO_3^-) – it is a detergent. The answer is (a).

19.(2020:22) Soaps form precipitates with calcium and magnesium ions dissolved in water. To avoid this problem, we use detergents because they do not form precipitates. The answer is (b).

20.(2020:23) In soap, for example, sodium stearate, the stearate ion is negatively charged. The long carbon chain is attracted to the grease by dispersion forces to form a micelle leaving the ion $-COO^-$ exposed to interact with water molecules. The answer is (c).

Chapter 6: Science Enquiry Skills

1.(2016 SP:11-b) Systematic error is caused by faulty equipment or faulty methodology and cannot be reduced by repeating an experiment. A pipette offers more sig figs and will lower random error. Answer is 'b'.

2.(2016:06-d) A systematic error can be identified by data that is precise (many values close together) but wrong. The answer is 'd'.

3.(2017:22-d) The concentration of $HC\ell$ is varied and the effect on the other variables are measured. This makes it the independent variable. The answer is 'd'.

4.(2017:23-c) The data involved numbers (quantitative) and was collected by the experimenter (thus it was primary data). The answer is 'c'.

5.(2017:24-a) The data was consistently wrong which means it was not random error. It was systematic caused by a wrong technique or poor equipment. The answer is 'a'.

6.(2017:25-d) The technique would mean there were differing methods used and the data would be less reliable. The answer is 'd'.